St. Francis Institute of Technology

(Engineering College)

An Autonomous Institute, Affiliated to University of Mumbai

NAAC A+ Accredited | CMPN, EXTC, INFT NBA Accredited | ISO 9001:2015 Certified

Department of Artificial Intelligence and Machine Learning

Academic Year: 2025-2026
Class / Branch: SE - AIML
Course: L303-Artificial Intelligence Lab.
Date of Assignment: 28/08/2025

Term: Odd (Jul 2025 - Dec 2025)
Semester: III
Lab No.: 5
Date of Submission: 15/09/2025

Lab-5 Recitation

Topic: First Order Logic

Q-1: Translate the following Propositional Logic to English sentences.
Let:

- E=Liron is eating
- H=Liron is hungry

(a) E => ¬H
(b) E ∧ ¬H
(c) ¬(H => ¬E)

Q-2: Translate the following English sentences to Propositional Logic.
Propositions: (R)aining, Liron is (S)ick, Liron is (H)ungry, Liron is (HA)appy, Liron owns a (C)at, Liron owns a (D)og
(a) It is raining if and only if Liron is sick
(b) If Liron is sick then it is raining, and vice versa
(c) It is raining is equivalent to Liron is sick
(d) Liron is hungry but happy
(e) Liron either owns a cat or a dog

Q-3: Write Truth tables of following propositions. (Write python code in Lab-5)
(a) (Rain → Wet) ∧ Rain
(b) (Study ∧ Homework) → Pass
(c) Light ↔ Switch

Q-4: Using Truth table, check whether following propositions are tautologies, contradictions pr contingencies and justify why? (Write python code in Lab-5)
(If all the values in the conclusion (final column) of a truth table are True, the formula is a tautology. If all the values are False, it is a contradiction. Otherwise, it is a contingency.)
(a) Three is a prime number.
(b) P
(c) P → P
(d) (P→ P) → P
(e) P → (P → P)
(f) (P∧Q) ∧ (¬ P ∨¬ Q)

Q-5: Which of the following propositions are **equivalent** in the sense that one can always be substituted for the other one in any proposition without changing its truth value? Why?

(a) First proposition: ¬ P second proposition: P ⇒ False
(b) First proposition: ¬ P second proposition: ¬P ∨ Q

Q-6: Which of the following formula is satisfiable? (Write python code in Lab-5)
(Satisfiable: A propositional logic formula is **satisfiable** if there exists **at least one assignment of truth values** to its variables that makes the formula **True**.)

(a) The student passed the exam or the student studied hard
(b) Either the switch is on or the bulb is glowing, but not both

Q-7: Verify entailment of a query from a knowledge base (KB) using truth tables. (Write python code in Lab5)

Example	**(KB)**	Query
1 (solved below)	P⇒Q, P	Q
2	P∨Q	¬P
3	R⇒S, *S⇒T, R*	T
4	R ∨ S	R
5	P⇒Q	P
6	(A∧B) ⇒C	B

<u>Solved Example-1 for reference:</u>
(Entailment: A knowledge base KB **entails** a formula α (written KB⊨α) if **every truth assignment that makes KB true also makes α true**.

- In other words: **α logically follows from KB**.
- If even **one assignment** makes KB true but α false → α is **not entailed** by KB.

Refer to solved example below)

Knowledge Base (KB) has two clauses:

1. P⇒Q
2. P

Query: Q
Ans:

P	**Q**	**P ⇒ Q**	**KB: (P ⇒ Q) ∧ P**	**α=Q True?**
T	**T**	T	**T**	**T**
T	F	F	F	F
F	T	T	F	T
F	F	T	F	F

Yes KB ⊨ Q (KB entails Q) because in the first row for P = True, Q = True, KB is True and also query Q is True. There is no assignment where when KB is True, Q is False.

Note: Attach the solved recitation with Lab experiment-5 in Journal.

Question 1 - Solution

Code:

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>To-Do List App</title>
    <style>
        * {
            margin: 0;
            padding: 0;
            box-sizing: border-box;
        }
        body {
            font-family: 'Segoe UI', Tahoma, Geneva, Verdana, sans-serif;
            background: linear-gradient(135deg, #0093E9 0%, #80D0C7 100%);
            min-height: 100vh;
            display: flex;
            justify-content: center;
            align-items: center;
            padding: 20px;
        }
        .container {
            background: white;
            padding: 40px;
            border-radius: 20px;
            box-shadow: 0 20px 60px rgba(0,0,0,0.2);
            width: 450px;
            max-width: 100%;
        }
        h1 {
            text-align: center;
            color: #2c3e50;
            margin-bottom: 10px;
            font-size: 2.5em;
        }
        .subtitle {
            text-align: center;
            color: #7f8c8d;
            margin-bottom: 30px;
            font-size: 1.1em;
        }
        .input-group {
            display: flex;
            gap: 12px;
            margin-bottom: 25px;
        }
        #taskInput {
            flex: 1;
            padding: 15px;
            border: 3px solid #0093E9;
            border-radius: 12px;
            font-size: 1.1em;
            transition: all 0.3s ease;
        }
        #taskInput:focus {
            border-color: #80D0C7;
            outline: none;
            box-shadow: 0 0 20px rgba(0, 147, 233, 0.2);
        }
        #addBtn {
```